

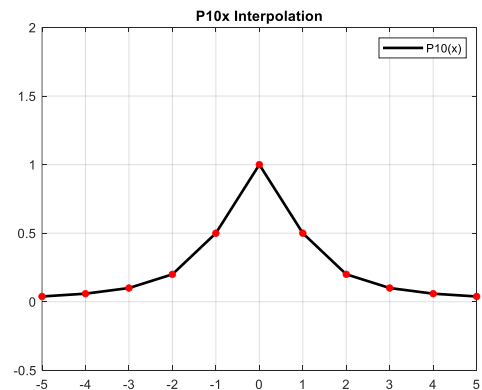
Written Problem 1)
Finding the Error using Theorem

[illegible]

Computer Problem 1) Runge Function Interpolations

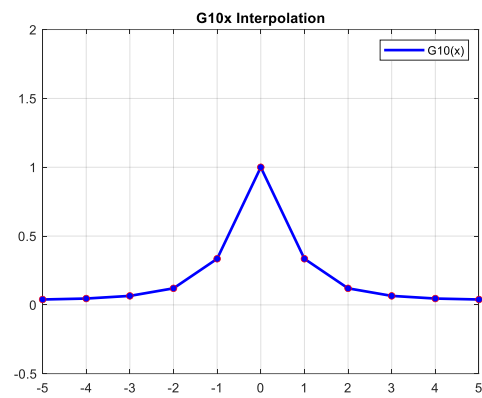
A) Plot $P_{10}(x)$ using equally spaced points on $[-5, 5]$

-Observations: the Plot of $P_{10}(x)$ are equally spaced and Symmetric. The plot of the node are exact values of the Runge Function. Is useless in a sense but the fine mesh graph will be a better visual (see part C below).



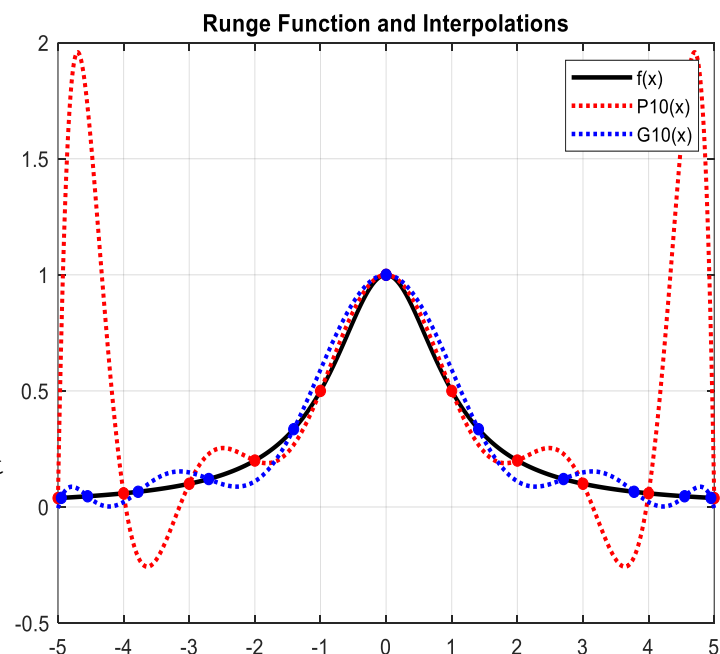
B) Plot $G_{10}(x)$ using equally spaced points on $[-5, 5]$

-Observations: This plot using the ChebyShev nodes that are equally spaced show a similar plot with the node being exact values of the Runge Function. The biggest difference is in the middle as this plot of Nodes create a steeper plot near the center. Once again the fine mesh will be a better visual.



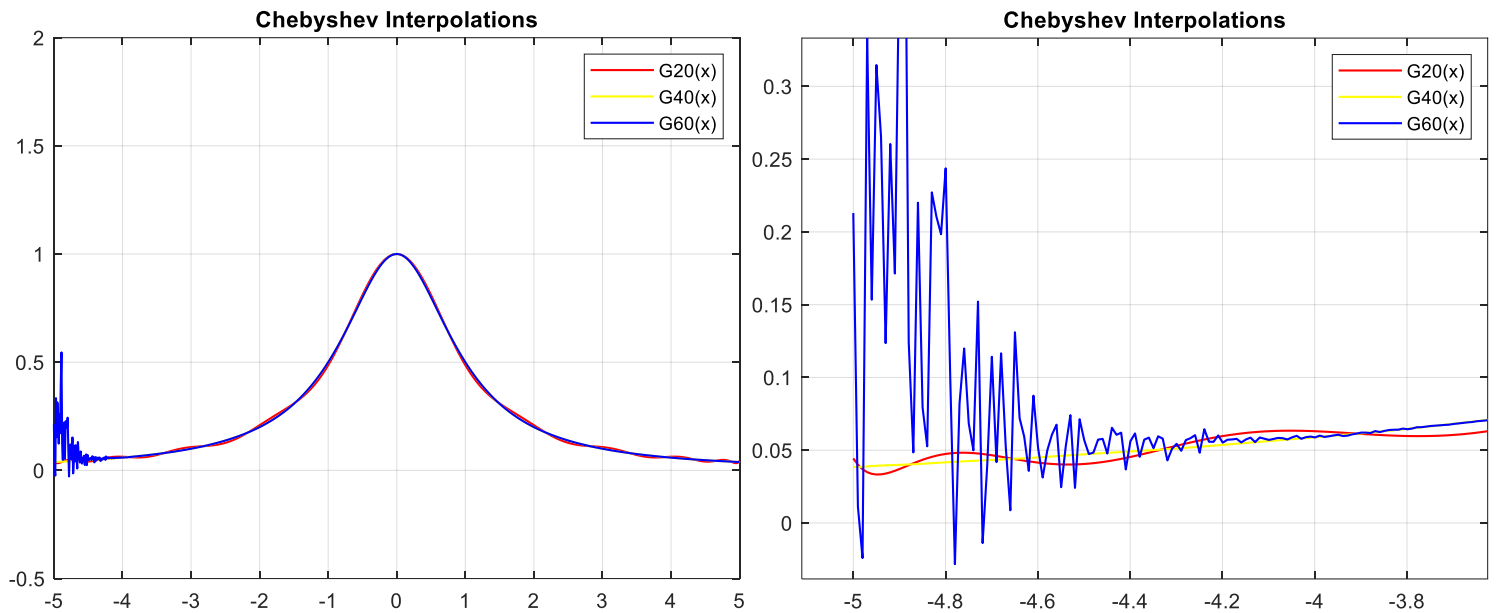
C) Using a Fine Mesh, Plot $f(x)$ and the Interpolations Above

-Observations: As promised the Fine Mesh shows the variation in the Oscillations between using ChebySheb nodes and a normal Interpolating Polynomial. The Nodes are also plotted and show that they are the exact values of the Runge Function. ChebyShev nodes work because the Points are concentrated near the



endpoints leading to less oscillation. The error is also less because of the product used at the end of the formula for finding the node, as that product is minimized thus minimizing the error.

D) Plotting Interpolations using Chebyshev nodes at different Degrees



Observations: The Higher Degree Nodes don't seem to improve the general accuracy of when using the Chebyshev nodes of degree 20, 40 and 60. The Interpolation Error Theorem from Written Problem #1 would suggest that the Error is small as the degree gets higher since the last term " $(b-1/n)^{n+1}$ " would lead to an increase in accuracy. Considering the Error Theorem, the Blue Line that represent the degree 60 oscillation suggest that there is an optimal degree that minimizes the error. This can be tested by changing the code and this plot suggest that the degree is somewhere between 40 and 60. Directions say to provide an educated guess and I assume that is the optimal degree, which I already stated is on the interval (40, 60).